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ORGAN TOXICITY AND MECHANISMS

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Short-term and long-term in vivo exposure to an ephedra- and caffeine-containing metabolic nutrition system does not induce cardiotoxicity in B6C3F1 mice

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Abstract Although conventional biomedical research has largely focused on mechanisms of weight loss and genetic aspects of obesity, most medical solutions are plagued by side-effects and fraught with complex questions. As a consequence, consumers are seriously considering herbal products, nutraceuticals and functional foods as alternatives to conventional medications. This is evidently driven by a growing consumer understanding of diet/disease links, aging-related consequences, rising health care costs, and advances in food technology and nutrition. This study investigated the effects of up to 12 months exposure to a multinutrient and botanical extract supplement (Metabolic Nutrition System Orange (MNSO) - sold by AdvoCare, Carrollton, TX, USA) at five dietary concentrations on serum biochemistry and target organ histopathology of the hearts of B6C3F1 mice. The MNSO is a unique combination of vitamins, minerals, omega-3 fatty acids and herbal extracts designed to provide a strong foundation of nutritional support, and to enhance thermogenesis and perception of energy. The MNSO contain extracts of citrus, ephedra, guarana, gingko, green tea and Ocimum. In this study, female B6C3F1 mice were fed control (−MNSO) or MNSO (one time to ten times, one time = daily human dose) diets. Animals were sacrificed after 4, 8 and 12 months', at which time blood was collected for serum chemistry analysis, and hearts were prepared for histopathology and tissue biochemistry. Food consumption and body weight changes were also monitored throughout the study. The MNSO exposure did not significantly affect any of the cardiosensitive enzymes [including creatine kinase (CK), lactate dehydrogenase (LDH) and aspartate aminotransferase (AST)] and normal histopathological architecture of the heart was observed. Although animals given the MNSO diet consumed more food, they were relatively leaner and more active compared to controls. The results indicate that ingestion of ephedra and caffeine for one year in the doses used as part of a comprehensive metabolic nutrition system does not significantly alter normal serum chemistry or induce any irreversible histological changes in the mouse heart, since this study employed up to ten times the normal human consumption dose of ephedra and the metabolic nutrition system.

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Introduction

At the beginning of the twenty-first century, a retrospective history of healthcare is likely to discuss the tremendous strides made by the pharmaceutical industry in treating, managing and curing human diseases. However, in recent years, the health-care industry has witnessed unprecedented public popularity in wellness, nutritional and alternative medical therapies. This shift in interest is partly due to the general emphasis of the "health-care system" on "disease state management" rather than "disease prevention." The cost of pharmaceuticals is an additional factor driving many consumers to find alternative therapies.

Americans alone spent an average of 20 billion dollars per year for the past five years on over-the-counter nutraceuticals and dietary supplements (http://www.theinfoshop.com/study/fd10449_nutraceuticals.html). This increased popularity of nutraceuticals and natural products is due to: (I) greater factual knowledge of specific diseases and disorders of the general population, (ii) availability and accessibility of products without a prescription, (iii) outcomes backed by increasingly scientific research observations and some clinical trials and (iv) the relative lack of stringent federal regulations. The Dietary Supplement Health and Education Act (DSHEA), enacted in 1994, allowed greater access to dietary supplements and separated supplements from drugs, which are regulated by the FDA. Because of the lesser regulation of dietary supplements, few or no studies exist regarding standardization of ingredients, daily human requirements, possible harmful effects, ability to interact with drugs and permissibility to consume under certain medical conditions.

Ephedra, also well-known as "ma huang," is an herb that has been widely used in dietary supplements for weight loss (Dulloo and Miller 1987; White et al 1997; Dulloo 2002; Boozer et al 2001, 2002). For thousands of years ephedra has been used in China and other Asian countries as an herbal remedy for asthma, cough, headache, fever, allergies and the common cold. Ephedrine alkaloids occur naturally as three diastereomeric pairs grouped by primary, secondary or tertiary amine functions. The D- and L-forms of synephrine both occur in nature, but the L-form predominates (Niemann and Gay 2003). Astrup et al reported in 1990 that a combination of ephedra and caffeine produced a significant loss of body weight and body fat compared to either ingredient alone or a placebo in healthy, overweight adults. Adverse events were mild and transitory. Their findings led to ephedra/caffeine mixtures being widely prescribed for obesity in Denmark, the country of origin of the study. In spite of its widespread use in the United States, few long-term controlled studies have examined the safety of ephedra. While a number of human studies have found ephedra/caffeine mixtures to be relatively safe with few side effects (Astrup et al 1992; Toubro et al 1993; Boozer et al 2002), others have suggested that these mixtures increase the risk of serious health problems (Zahn et al 1999; Haller and Benowitz 2000; Morgenstern et al 2003). Bent et al (2003) have suggested that ephedra is associated with an increased risk of adverse events in humans as compared to other herbs based on poison Control Center toxicity events surveillance data. The Rand Health Study (RAND 2003) concerning the safety of ephedra and ephedrine suggested a relationship between use of ephedra or ephedrine and serious adverse events, although the available evidence was not statistically significant. Other than the early studies by Chen and Schmidt (1924) and Chen (1926a, 1926b), no studies can be found in the literature on the subchronic or chronic effects of ephedra or ephedra in combination with caffeine on multiple vital organs in humans or animals. The current interest in natural herbal remedies has led to a resurgence in the number of products containing ephedra. A recent meta-analysis raised further concerns about potential serious adverse events from ephedra and ephedra/caffeine mixtures (Shekelle et al 2003). Based on the meta-analysis as well as massive negative publicity, the U.S. Food and Drug Administration recently issued a ruling banning the sale of ephedra/caffeine mixtures for weight loss and athletic performance (FDA 2004).

Prior to the FDA ban on sales of ephedra, we investigated long-term exposure effects of a metabolic nutrition system (MNS Orange, AdvoCare) containing a unique mixture of nutrients and phytochemicals on multiple vital target organs of B6C3F1 mice. Since the majority of the reported toxic effects of ephedra have been cardiovascular-related, this investigation employed acute, subchronic and chronic MNSO-exposure modes to explore cardiac changes in mice. This study employed not only a human-equivalent dose (one time) but also two- to tenfold higher doses (two, three, six and ten times) for up to 12 months to assess biochemical and histopathological changes in the mouse heart.

Materials and methods

Animals and animal housing conditions

All experiments were conducted in accordance with GLP standards. Four-week old female B6C3F1 mice were obtained from NCI, Frederick, MD and given access to lab chow (Purina Laboratory rodent chow, St. Louis, MO, USA) and tap water ad libitum. All animals were allowed to acclimate in an environment of controlled temperature (22–25 °C), humidity and light/dark cycle in the LIU animal facility for weeks prior to study. All cages were examined several times every day to identify abnormalities, health conditions, sick or dead animals. Body weights of the animals were recorded once every two weeks to monitor growth. Cages were changed twice a week to provide clean housing. All animal procedures received prior approval of the Institutional Animal Care and Use Committee and met or exceeded current local, state and federal standards.

Animal treatments

Animals in this study were divided into six groups. Experiments began with 50 animals in each treatment group:

Group 1 Control; received regular chow.

Group 2 One time MNSO diet. The HPLC analysis of the MNSO diet prior to formulating the chow indicated that the human equivalent dose contained 40 mg ephedra and 150 mg caffeine. Received chow premixed with one time the human equivalent of MNSO.

Group 3 Two times MNSO diet. Received chow premixed with two times the human equivalent of MNSO.

Group 4 Three times MNSO diet. Received chow premixed with three times the human equivalent of MNSO.

Group 5 Six times MNSO diet. Received chow premixed with six times the human equivalent of MNSO.

Group 6 Ten times MNSO diet. Received chow premixed with ten times the human equivalent of MNSO.

The MNSO exposure levels were achieved as follows. The MNSO in powdered form was supplied to the diet manufacturer. Pre-calculated amounts of MNSO powder (40 mg ephedra in 13.26 g MNSO mix) were mixed with Purina base diet powder (Rodent diet #5001) in order to achieve an average daily intake of 1.2 mg ephedra per mouse per day (with an assumption that a 30 g mouse eats 5 g of chow per day). The human equivalent one time daily dose was calculated based on a 70 kg human consuming 40 mg ephedra/day. Control and MNSO-mixed base diet powders (Purina #5001 base) were repelleted in identical form (0.5″. pellet) and supplied to our facility, which was provided to the animals ad libitum. Pellets were stored at 4–8 °C as recommended by the manufacturer of the chow and MNSO system. This diet provided continuous MNSO exposure during each meal throughout the experimental period.

Base rodent chow consisted of: (I) crude protein not less than 23%, (ii) crude fat not less than 4.5%, (iii) crude fiber not more than 6%, (iv) ash not more than 8% and (v) minerals not more than 2.5%. This diet is a Constant Nutrition formulation recommended for mice and several other rodents. The constant formula feature is designed to minimize nutritional variables in long-term studies. It is formulated for life-cycle nutrition. This product has been the standard of biomedical research for approximately 50 years. The MNSO powder supplied by AdvoCare International was mixed with this diet.

The MNSO contained a total of 1 g of fat (in a serving size of 13.26 g). The recommended intake used to be four packets (two white packets + two silver packets) per day. The breakdown of ingredients of white and silver packets is indicated in Table 1. Overall, this product has 40 mg of ephedrine group alkaloids per serving in the form of herbal extracts, as well as 150 mg of caffeine alkaloids per serving.

Randomly-sorted animals ($n = 9–12$ per group) were sacrificed by decapitation at four-month intervals. Blood was collected (pooled from three to four animals to make one sample) for serum chemistry analysis to indirectly monitor the functioning of the vital organs. Blood was allowed to coagulate, centrifuged at 14,000 rpm (at 15 °C in a table-top small centrifuge) to separate the serum and all analyses were completed within 72 h after collection. Samples were stored at −70 °C prior to analysis, if necessary. Mouse hearts were collected, sectioned, and the tissues were preserved in 10% buffered formalin for histopathology. Tissue sections were processed by a histopathology laboratory highly experienced in FDA/EPA GLP studies. The H&E-stained heart sections were examined with a Carl-Zeiss brightfield microscope.

Serum chemistry

Serum chemistry analysis was performed at a State- and FDA-approved certified diagnostic laboratory used to perform clinical chemistry, including for humans. Serum samples were collected and kept frozen at −70 °C for delivery in liquid nitrogen to the diagnostics laboratory (Bio-Medical Laboratories, East Brunswick, NJ, USA). An advanced COBAS MIRA (Roche Diagnostics, IN, USA) was employed for the assays (protein, lipid, carbohydrate, electrolyte and enzyme profiles) using reagents from Diagnostic Chemicals, Ltd. Serum samples were pooled from three to four animals and data were analyzed at least in triplicate (9–12 animals per treatment). Serum chemistry values were compared with appropriate standards provided by the manufacturer.

Histopathological evaluation

Heart sections preserved in 10% phosphate buffered formalin were sent to IDEXX Laboratories (San Jose, CA, USA) for further processing, paraffin embedding, sectioning and H&E staining. Stained sections were examined under a microscope for normal, apoptotic, apocrotic and necrotic cells (Ray et al 1996; Ray 1999). Morphological changes were interpreted based on minimum 400× or 1000× magnifications. Sections were imaged with VIDEO microscopy for any type of abnormality. A board-certified histopathologist (veterinarian) read all the final H&E stained slides, interpreted findings and provided a report (GLP spirit). One of us (SDR) examined all of the slides and verified the report for accuracy. Photographs were taken using a Carl-Zeiss brightfield microscope (at 100×). The entire section was examined for evaluation. A minimum of two sections/heart and three hearts/three animals were evaluated for interpretations.

Statistical analysis

Results are presented as mean ± SEM unless otherwise indicated. Data were analyzed for significance ($P < 0.05$) using analysis of variance (ANOVA) followed by Fisher PLSD test (Stat View II for Macintosh, Version 1.04, Abacus Concepts Inc., Berkeley, CA, USA) or evaluated by linear regression analysis and correlation. Differences were attributed to treatment rather than chance variance when $P < 0.05$.

Table 1 Composition of MNS Orange

Serving size (13.26 g)	White packets		Silver packets	
	Amount per serving (1)	Percent DV	Amount per serving (2)	Percent DV
Calories	9			
Calories from fat	9			
Total fat	1 g	2		
Vitamin A (as palmitate)	2500 IU	50		
Vitamin A (as beta-carotene)	12500 IU	250	2000 IU	40
Vitamin C (as ascorbic acid/mineral ascorbates)	600 mg	1000	600 mg	1000
Vitamin D (as ergocalciferol)	400 IU	100		
Vitamin E (as D-alpha tocopheryl succinate)	150 IU	500		
Thiamine	9 mg	600		
Riboflavin	10.2 mg	600		
Niacin/Niacinamide	120 mg	600	40 mg	200
Vitamin B6 (as pyridoxine HCl)	12 mg	600	4 mg	200
Folic acid	800 μg	200		
Vitamin B12 (as cyanocobalamin) 36 μg	600%			
Biotin	300 μg	100		
Pantothenic acid	40 mg	400		
Calcium (as amino acid chelate) 150 mg	15%			
Phosphorus (as amino acid chelate) 100 mg	10%			
Iodine (from kelp)	150 μg	100	50 μg	30
Magnesium (as amino acid chelate/phosphate)	200 mg	50	100 mg	30
Zinc (as monomethionine)	15 mg	100	2 mg	12
Selenium (as selenomethionine) 80 μg	115%			
Copper (as amino acid chelate)	2 mg	100		
Manganese (as amino acid chelate)	4 mg	200		
Chromium (as polynicotinate)	100 μg	80	100 μg	80
Molybdenum (as amino acid chelate)	50 μg	70		
Potassium (as amino acid chelate/phosphate)	100 mg	2	200 mg	4
Boron (as amino acid chelate)	300 μg	[a]		
Vanadium (as bis-maltolato-oxovanadium)	50 μg	[a]	200 μg	[a]
Silicon (as amino acid chelate)	500 μg	[a]		
Inositol	6 mg	[a]		
Choline (as bitartrate)	60 mg	[a]		
Coenzyme Q10	150 μg	[a]		
Octacosanol	2000 μg	[a]		
Ribonucleic acid (RNA)	2000 μg	[a]		
Odorless garlic powder (*Allium sativum*)	50 mg	[a]		
L-Glutathione	5 mg	[a]		
Gamma-oryzanol	5 mg	[a]		
Citrus bioflavonoids	100 mg	[a]		
Red wine polyphenols	5 mg	[a]		
Milk thistle extract (fruit: *Silybum marianum*)	5 mg	[a]		
Ginkgo extract (leaf: *Ginkgo biloba*)	10 mg	[a]		
L-Methionine	100 mg	[a]		
Eicosapentaenoic acid (from marine lipids)	180 mg	[a]		
Docosahexaenoic acid (from marine lipids)	120 mg	[a]		
Garcinia extract (fruit: *Garcinia cambogia*)			2000 mg	[a]
Tulsi extract (leaf: *Ocimum sanctum*)			50 mg	[a]
Taurine			50 mg	[a]
L-Carnitine			25 mg	[a]
Beta-sitosterol			25 mg	[a]
Gymnema extract (leaf: *Gymnema sylvestre*)			10 mg	[a]
Ephedra extract (leaf/stem: *Ephedra sinica*)			500 mg	[a]
Guarana extract (seed: *Paullinia cupana*)			550 mg	[a]
Green tea extract (leaf: *Camellia sinensis*)			20 mg	[a]
Oolong tea extract (leaf: *Camellia sinensis*)			200 mg	[a]
Eleuthero extract (root: *Eleutherococcus senticosus*)			50 mg	[a]

[a]No % daily value (DV) established for this ingredient. % daily values are based on a 2000 calorie diet

Results

Figure 1 shows the food consumption patterns of the animals over a period of 12 months. Human equivalent consumption dose, 40 mg of ephedra or one time concentration did not alter the food consumption pattern significantly compared to controls. The fluctuations in food consumption observed are normal for rodents during their growth cycle. The overall consumption patterns show that animals on the MNSO diets generally

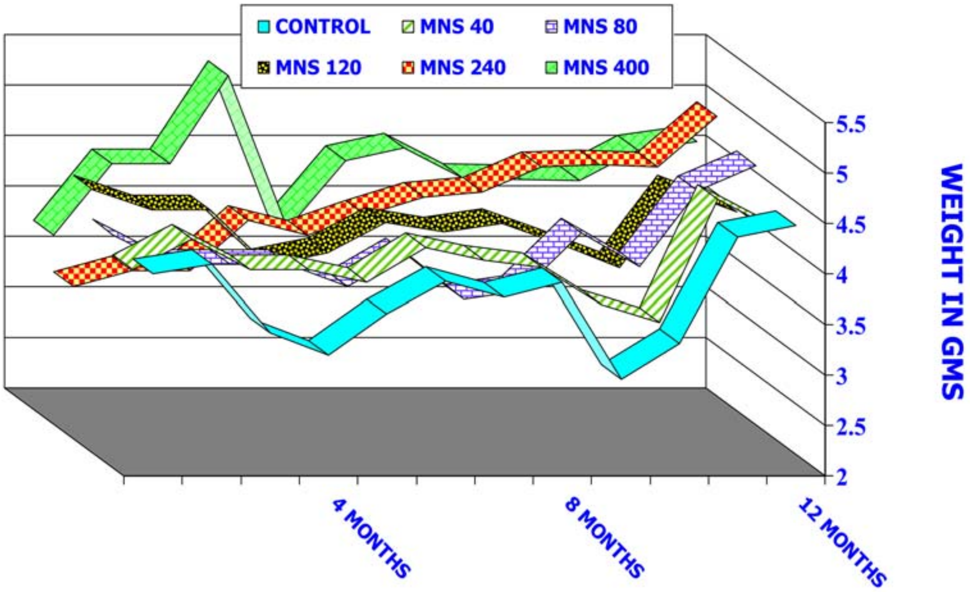

Fig. 1 Average monthly food consumption patterns of female B6C3F1 mice kept on control (−MNSO) and MNSO diets. Six to eight animals were housed per cage. A weighed amount of chow was provided and the consumption was monitored on a regular basis. Experiments began with 50 animals in each group. Data represents average weight of the remainder of the animals after every four months. Final data were calculated on a per animal basis

consumed more, with consumption of six and ten times being the two highest.

Weight gain patterns are presented in Fig. 2. Unlike the food consumption pattern, little fluctuation was noted in growth patterns during the entire study. Human consumption dose (40 mg of ephedra, one time) consistently maintained normal physical appearance, minimized the weight gain, and followed a normal growth curve relative to controls throughout the study. Animals maintained on the two and three times diets did not show any significant change in their body weights compared to controls. However, the two diets with the highest concentrations, six and ten times, showed significantly lower body weights compared to controls. This reduction in the rate of weight gain was evident after eight months of exposure and until the end of the study. Although growth patterns did not follow a strict dose response effect, six and ten times concentrations showed a dose-dependency. The MNSO exposure clearly helped to maintain a normal physical appearance without any significant weight gain.

Overall physical appearance of the animals after 4 and 12 months of MNSO exposure (one, three, six and ten times) is presented in Figs. 3 and 4, respectively. Control animals generally had a bulkier appearance compared to the MNSO-exposed animals. The leaner appearance of the MNSO groups was noticeable after four months and persisted throughout the study. Con-

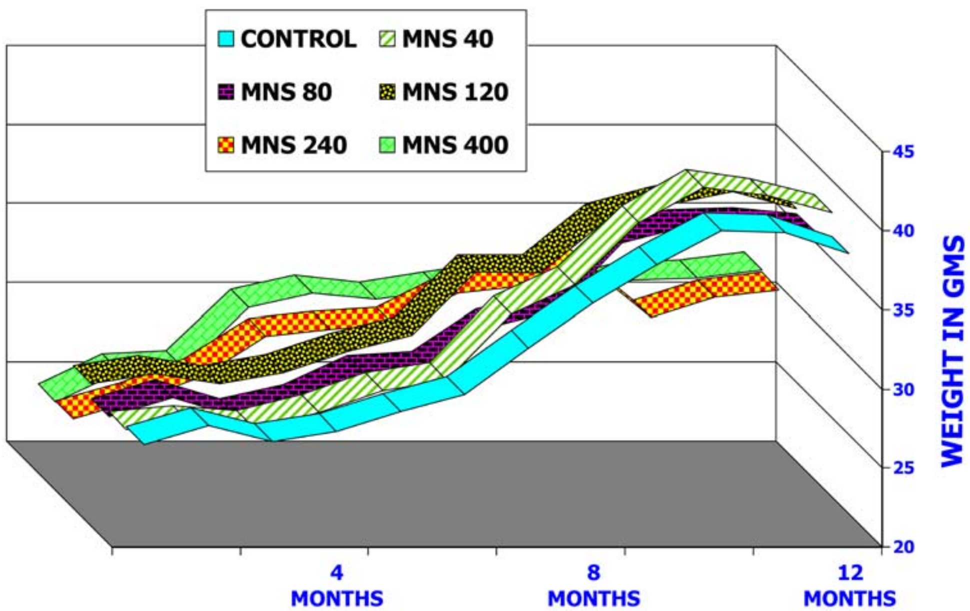

Fig. 2 Mean body weights of female B6C3F1 mice fed on control (−MNSO) and MNSO diets. Six to eight animals were housed per cage. Body weight gain was recorded every week until the end of the study. Experiments began with 50 animals in each group. Data represents average weight of the remainder of the animals after every four months

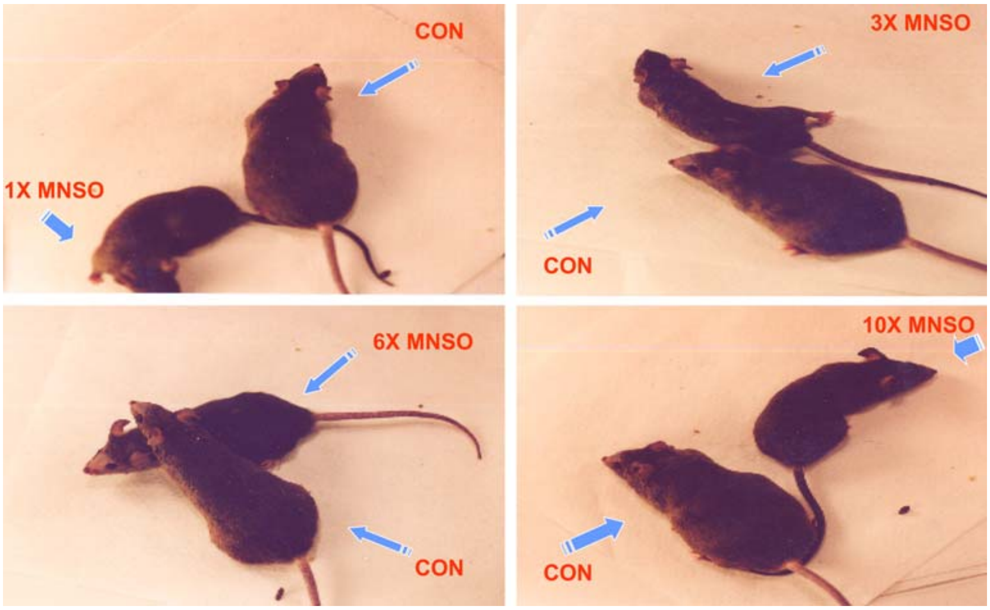

Fig. 3 Effects of four months of exposure to control (−MNSO) and MNSO diets on the overall physical appearance of representative female B6C3F1 mice

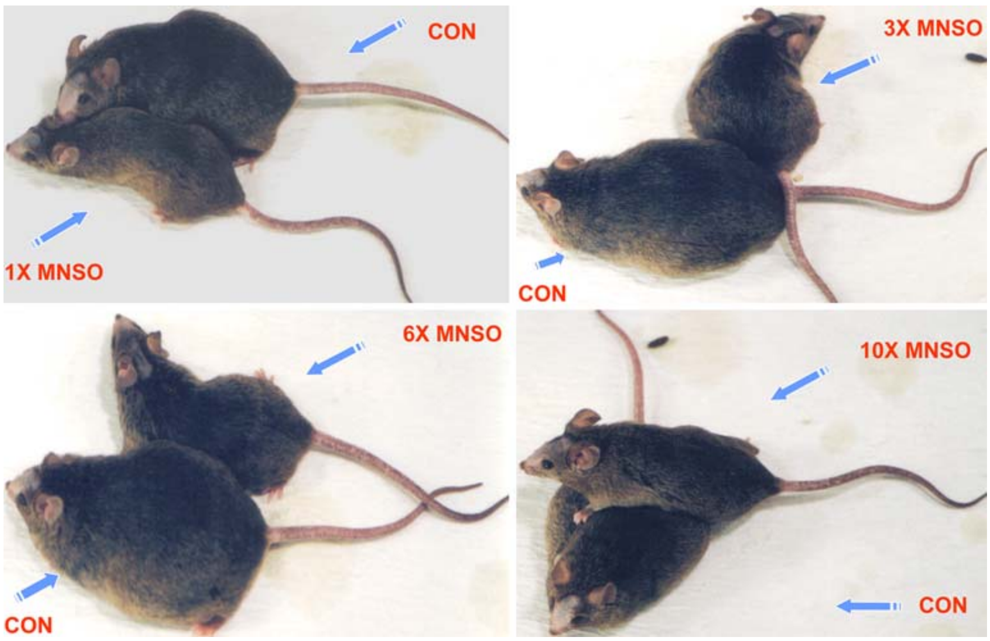

Fig. 4 Effects of 12 months of exposure to control (−MNSO) and MNSO diets on the overall physical appearance of representative female B6C3F1 mice

siderable differences in the overall physical appearance (Figs. 3, 4) were consistent with weight loss (Fig. 2) observed at the six and ten times' MNSO concentrations. Although the actual body weights did not show a dose-response at the lower MNSO concentrations (one to three times), a dose-dependent effect was clearly apparent in the appearances of the animals, indicating lower body fat. All doses of MNSO were effective in minimizing weight gain and maintaining normal weight of the animals.

Figure 5 shows MNSO-induced changes in creatine kinase activity (CK) in serum. This enzyme is considered the most sensitive and dependable cardioactive marker. The CK is a ''leakage'' enzyme present in high concentrations in the cytoplasm of cardiac myocytes and is the enzyme most widely used for evaluating neuromuscular disease. The CK is a dimeric molecule composed of two types of monomers, the ''M'' and ''B'' subunits. Routine assay measures total CK activity.

The CK activity gradually declined in control animals with increasing age. Control animals showed nearly a 2000 unit decrease after 12 months. All concentrations of MNSO modestly elevated the CK activity level at four months in a dose-dependent manner except for the highest dose (ten times), which was slightly inhibitory. At eight months, most doses of MNSO except the ten times dose showed near-normal enzyme activity. At the conclusion of the study, three lower MNSO concentra-

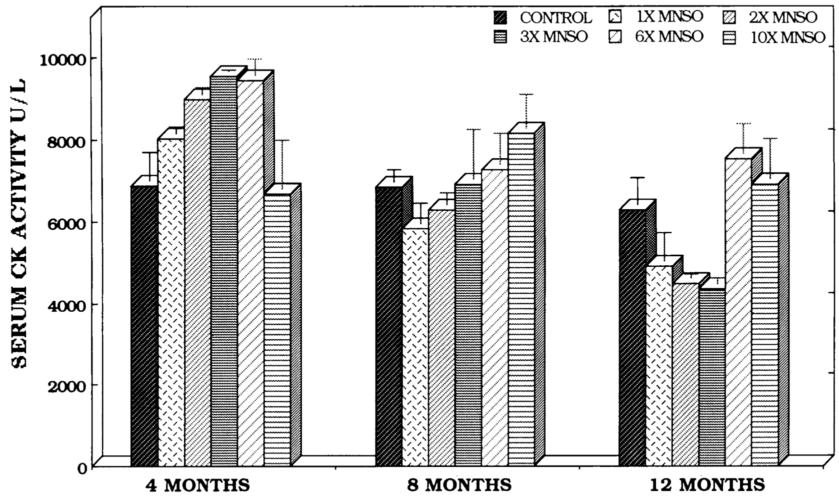

Fig. 5 Changes in serum creatine kinase (CK) activity for female B6C3F1 mice fed control (−MNSO) and MNSO diets for 4, 8 and 12 months. Sera from three to four mice were pooled to make one sample. *Bars* reflect at least $n = 3$

tions (one to three times) showed significantly declining activity and the two highest concentrations (six and ten times) exhibited small insignificant elevations. The overall pattern of this enzyme activity disclosed no serious adverse effects of MNSO on the heart. In general, the results indicate that the long-term administration of this diet may have had cardioprotective effects at the one to three times' doses, while the six and ten times' doses resulted in CK levels within the normal range.

Changes in lactate dehydrogenase (LDH) activity are presented in Fig. 6. This enzyme is found primarily in the cells of heart, lung and blood. Usually serum LDH activity reflects cardiac muscle condition and is used as a reliable cardiosensitive marker. Elevated LDH levels are usually associated with myocardial infarcts, liver disease, renal disease and progressive muscular dystrophy. Unlike serum CK activity (Fig. 5), serum LDH level did not change in control animals during the entire study (Fig. 6). Feeding MNSO for up to 12 months did not produce any major changes in serum LDH activity except reflecting a slightly higher level of activity of this enzyme. Minor fluctuations were noted with all concentrations of MNSO.

Changes in aspartate aminotransferase (AST, formerly known as SGOT) enzyme levels with time are presented in Fig. 7. The AST is an enzyme primarily located in the heart and liver. Therefore, this enzyme serves as an additional biomarker of cardiac function. Although AST level is not entirely reflective of cardiac malfunction, it serves as a reliable predictor for cardiotoxicity. The AST levels gradually increased as the control animals aged. Control animals showed AST activity of approximately 194 units at four months, 257 units at eight months and 292 units at 12 months. A direct correlation was observed between AST activity and the age of the animals. The lower MNSO concentrations (one to three times) were associated with a 50% increase in AST activity, whereas the two highest concentrations (six and ten times) produced a 50% decrease in enzyme activity compared to controls at four months. The AST levels returned to normal by eight months and beyond in MNSO-fed animals. Overall AST values at 8 and 12 months exhibited no adverse changes as a result of exposure to MNSO-containing diets. Only the six times' dose showed an abnormal increase in AST activity after 12 months of exposure.

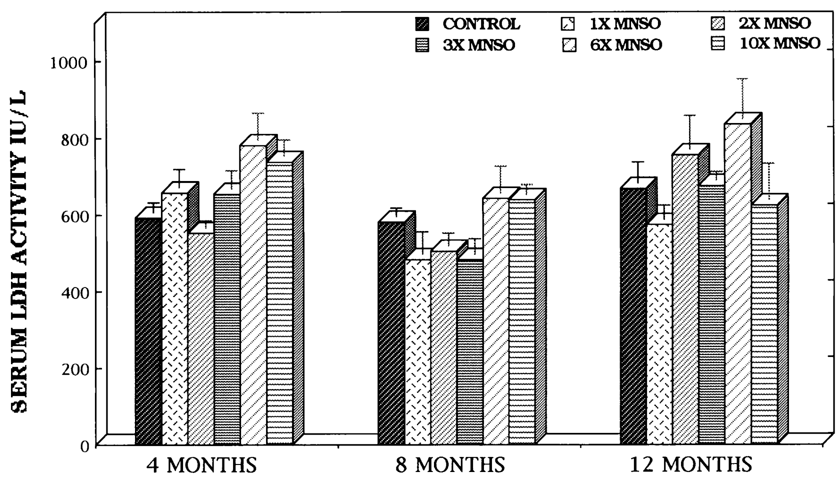

Fig. 6 Changes in serum lactate dehydrogenase (LDH) activity for female B6C3F1 mice fed control (−MNSO) and MNSO diets for 4, 8 and 12 months. Sera from three to four mice were pooled to make one sample. *Bars* reflect at least $n = 3$

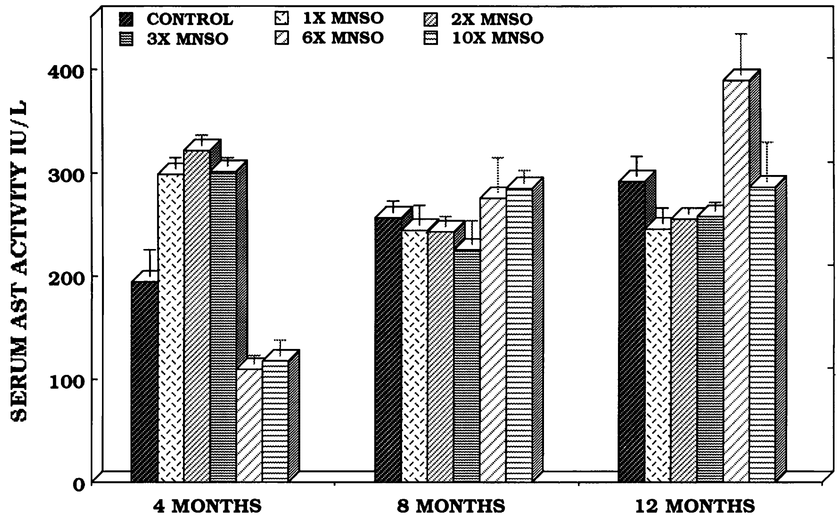

Fig. 7 Changes in serum aspartate aminotransferase activity (AST) for female B6C3F1 mice fed control (−MNSO) and MNSO diets for 4, 8 and 12 months. Sera from three to four animals were pooled to make one sample. *Bars* reflect at least $n = 3$

Histopathological changes induced after MNSO exposure are presented in Figs. 8, 9, 10. Serum chemistry typically reflects a combination of reversible and irreversible biochemical events, whereas histopathological alterations discern actual changes at the cellular and organ level. Three representative hearts from each treatment group and two sections from each heart were evaluated. Each figure shows one control and one section each from the hearts of animals receiving one, three and ten times MNSO concentrations in the diet. No differences in the color intensities were observed in the H&E-stained heart sections, mirroring an absence of any overt aging-related effects. In fact, myocyte organization in all the areas appeared identical in control sections as a function of age. Similarly, MNSO-exposed sections appeared identical to controls. Normal tissue architecture in MNSO-exposed tissues resembled control tissues. Histopathological results corroborated with the reversible serum chemistry changes presented in Figs. 5, 6, 7. Although H&E staining is not glycogen-sensitive, uniformity of the staining pattern in all MNSO-exposed tissues possibly reflected a lack of change in the glycogen levels.

Discussion

Few nutraceutical and dietary supplement products on the market are subject to safety and efficacy studies in animals and controlled clinical trials in humans. The present study assessed the long-term effects of a nutritionally-balanced formulation composed of essential

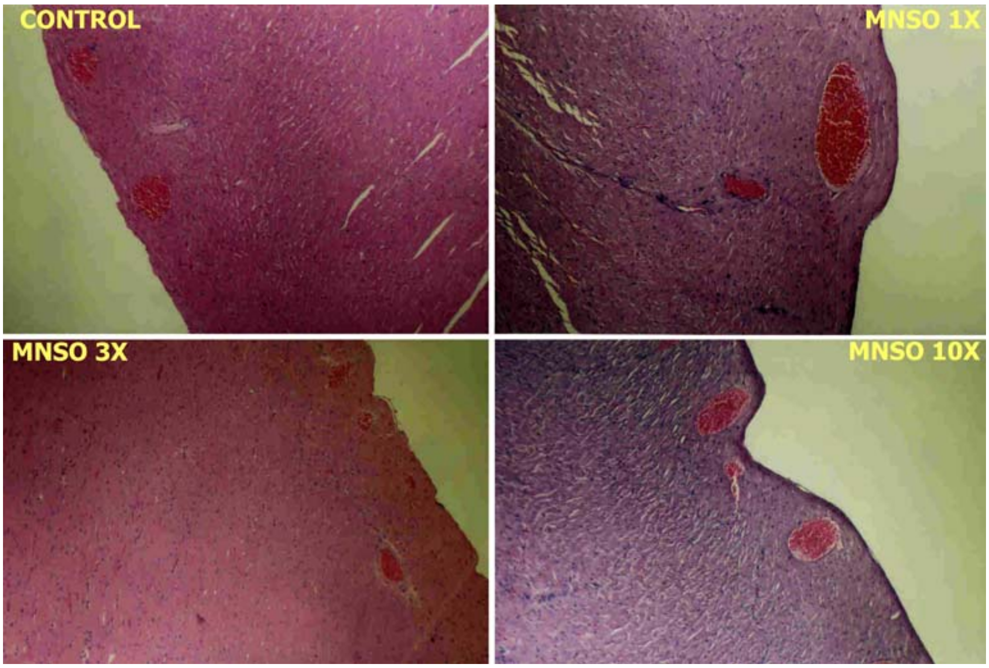

Fig. 8 Representative light photomicrographs (H&E-stained; 100×) showing normal cardiacmyocyte architecture (control), and effects of being fed various concentrations of MNSO (one, three and ten times) for four months on the female B6C3F1 mouse hearts. Staining intensities are identical in control and MNSO pre-exposed sections. No treatment-related histopathological abnormalities were observed

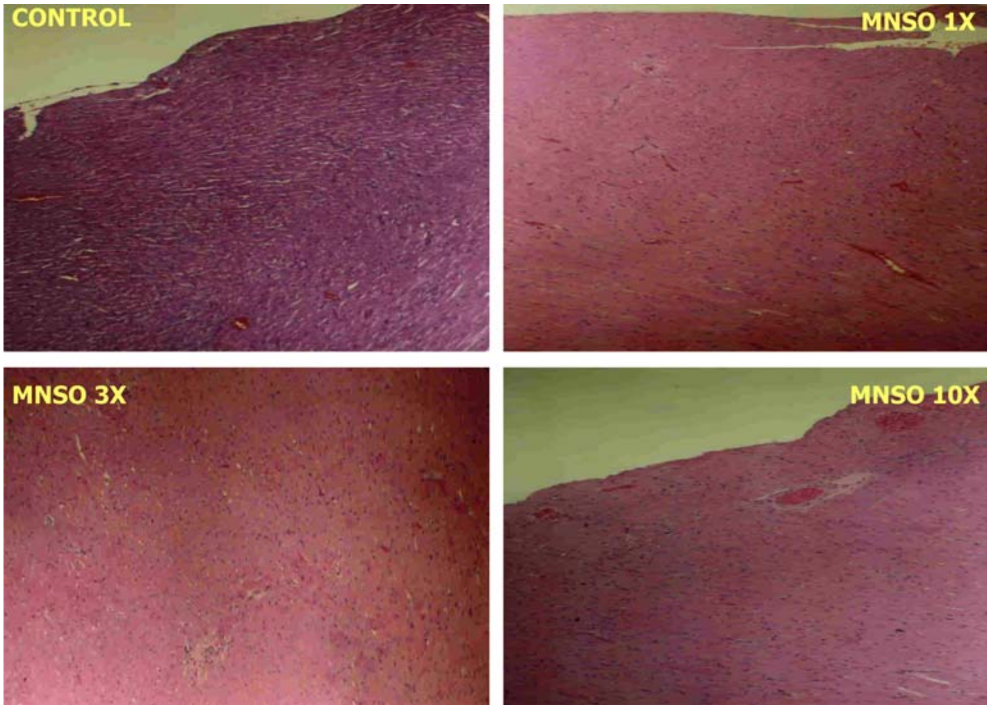

Fig. 9 Representative light photomicrographs (H&E-stained; 100×) showing normal cardiacmyocyte architecture (control), and effects of being fed various concentrations of MNSO (one, three and ten times) for eight months on the female B6C3F1 mouse hearts. Staining intensities are identical in control and MNSO pre-exposed sections. No treatment-related histopathological abnormalities were observed

vitamins, minerals, omega-3 fatty acids and phytochemicals known as MNSO. This product was fed to mice at up to tenfold the recommended human dose for up to 12 months. The MNSO also contained caffeine and ephedra. The findings in this study may represent the first systematic investigation on the effects of a caffeine- and ephedra-containing nutrition system. The nutritional system that tested up to tenfold the recommended human equivalent dose for one year did not produce any adverse cardiovascular effects in mice. Nine to 12 randomly chosen animals were used per group per treatment and sacrificed at 4, 8 and 12 months. Although treatment-induced fluctuations occurred in various treatments throughout the study, no major changes in serum CK, LDH or AST activities indicative of cardiovascular damage were observed. These results were confirmed based on histopathological analysis.

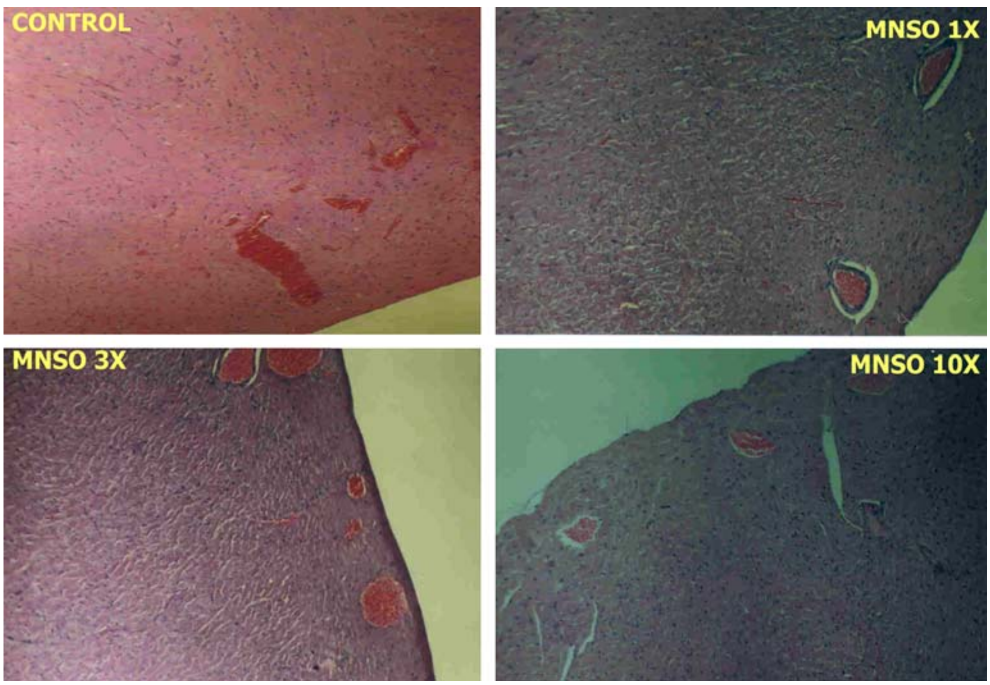

Fig. 10 Representative light photomicrographs (H&E-stained; 100×) showing normal cardiacmyocyte architecture (control), and effects of being fed various concentrations of MNSO (one, three and ten times) for 12 months on the female B6C3F1 mouse hearts. Staining intensities are identical in control and MNSO pre-exposed sections. No treatment-related histopathological abnormalities were observed

The most dramatic effect observed was the visible slimness of all animals at all doses of the MNSO-containing diets (Figs. 3, 4), and the significant loss of weight at the two highest (six and ten times) doses (Fig. 1). In general, control animals appeared heavier and fatter than treated animals. This conclusion was confirmed when internal organs were removed, and large amounts of fat was observed in the viscerae and the peritoneal cavities. Actual fat content per animal was not determined on a quantitative basis.

The reduced weight gain and weight loss in these animals are consistent with reports regarding the efficacy of ephedra/caffeine mixtures (Astrup et al 1991, 1992; Toubro et al 1993) and humans (Daly et al 1993; Boozer et al 2002). Several possible explanations exist for the physical appearance of the animals on the MNSO diet. The unique combination of ingredients in the MNSO diet may have discouraged fat accumulation while promoting protein anabolism. The MNSO may have promoted fat catabolism while favoring the building of muscle mass. The MNSO diet may have enhanced the basal metabolic rate, resulting in greater energy expenditure at the expense of fat, which has been reported by others (Astrup et al 1992; Zahn et al 1999). Finally, this unique metabolic system may have discouraged fat biosynthesis and accumulation. These possible pathways are speculative, but provide possible explanations for the ability of the MNSO dietary supplement to exert the observed effects. A combination of these proposed possibilities may also be operative. Reduced food intake has also been reported to occur in some investigations with exposure to ephedra/caffeine mixtures (Zahn et al 1999; Haller et al. 2000).

Overall, the MNSO did not induce vital target organ toxicity or serum chemistry anomalies at the doses that were used and the time points that were examined. Transient changes observed in some of the parameters did not translate into overt histopathological changes. Some of the changes induced by the MNSO diet could be viewed as mildly protective while other changes could be considered mildly detrimental. Overall, the data obtained from the three cardiosensitive enzymes (CK, LDH and AST) were not indicative of cardiotoxicity, even when used on a daily basis at up to tenfold the human equivalent daily dose for 12 months.

The histopathological observations (Figs. 8, 9, 10) corroborated the biochemical findings. None of the doses of MNSO at any stage of the study induced significant morphological changes. The normal myocyte architecture remained intact. Heart sections were also examined under higher magnifications for fine architectural changes (400× and 1000×). Cell deaths observed via necrosis and apoptosis for MNSO-treated animals were near-normal and the percent of these cells was lower than for control animals (data not shown). Under high magnification (1000×) branching muscle fibers, central nuclei with perinuclear (non-fibrillar) sarcoplasm, cross bands as in striated muscle and intercalated discs crossing the muscle fibers transversely including the *z*-bands were all normal. A thorough analysis of multiple sections did not disclose any MNSO-induced treatment effects. Overall, this caffeine-containing and ephedra-containing MNSO dietary supplement was found to be non-toxic at all doses over the 12 months of the study.

Nine to 12 animals were sacrificed from each dietary treatment group and the control group at 4, 8 and 12 months. At least ten animals were maintained for an additional six months (total of 18 months) thereafter on each of the MNSO-containing diets. Death of an occasional animal is common in an animal population of this size. During the first six months of exposure, one animal died in each of the groups that were given the two and six times' concentrations of MNSO, while two animals died in the ten times group. No animals died in any of the groups during the next 12 months (up to 18 months). Thus, none of the deaths could be attributed to MNSO. Deaths of the animals could be attributed to normal clash-related injuries followed by the cannibalistic behavior of the rodents.

Although the lay press has published numerous articles about the potential health effects of ephedra and anecdotal reports of ephedra toxicity, a robust body of literature exists regarding the safety and efficacy of ephedra, and ephedra in combination with caffeine in humans and animals.

The present study represents the first detailed, in-depth study on the effects of ephedra since the early work of Chen and Schmidt (1924) and Chen (1926a, 1926b) that were published approximately 80 years ago. The above results indicate that, when used in moderation and for prolonged periods of time in healthy animals, ephedra is non-toxic even at relatively high doses. Reasons for this apparent lack of toxicity may relate to the health of the animals, the controlled conditions under which the studies were conducted, and the fact that ephedra and caffeine were combined in a unique and complex dietary formula. Furthermore, it should be understood that since the amounts of ephedra and caffeine were increased in the diet, so also were all other ingredients in the MNSO formula. The combination of other ingredients may have exerted a modulatory effect.

There are a number of potential reasons why adverse events have been reported to occur among some users of ephedra-containing products (Molnar et al 2000; Haller et al 2002; Morgenstern et al 2003; Shekelle et al 2003). As with all xenobiotics, individual sensitivities vary. Furthermore, abuse and excessive consumption contrary to amounts generally recognized as safe and effective may have occurred. Some products containing ephedra have unwisely contained amounts greater than is recognized as safe (CRN 2002) and/or have recommended more frequent dosing than appropriate. Finally, some individuals with existing health conditions such as hypertension or cardiovascular diseases may have ignored contraindication labels on bottles of dietary supplements containing ephedra and experienced adverse events.

In summary, mice fed ephedra and caffeine in a complex metabolic nutrition system for up to 12 months with concentrations of ephedra as high as tenfold the human equivalent dose showed no abnormal changes in serum or histopathological markers of cardiovascular function.

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